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23 JUNE 1949
SUPERSEDING
JAN-M-382
28 August 1946

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

MAGNESIUM POWDER
(For Use in Ammunition)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for the use of procurement services of the respective Departments

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers powdered magnesium suitable for use in ammunition.

1.2 Classification. — Magnesium powder shall be furnished in the following types, grades, and granulations:

Type I.—Flaked and/or chip:

Grade A (96 percent minimum magnesium content)

Granulation No.:	Nominal mesh size [1]
1	40 to 80
2	40 to 80 (alternate).
3	50 to 100
4	50 to 100 (Army).
5	50 to 120
6	80 to 120
7	100.
8	120 to 200
9	140.
10	200.
11	80 to 200
12	120 to 200 (Army).

Grade B (86 percent minimum magnesium content)

13	20 to 50.

Type II.—Oblong chip with rounded edges (96 percent minimum magnesium content):

Granulation No.:	
14	50 to 100.

Type III.—Atomized (98 percent minimum magnesium content):

Granulation No.:	Nominal mesh size [1]
15	100 to 200.
16	200 to 325.
17	50 to 100.

[1] Nominal mesh sizes are based on U. S. Standard sieves in accordance with Specification RR-S-366.

2. APPLICABLE SPECIFICATIONS, AND OTHER PUBLICATIONS

2.1 Specifications.—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATION

RR-S-366—Sieves; Standard, Testing.

U. S. ARMY SPECIFICATIONS

50-0-1—General Specification for Ammunition Except Small Arms Ammunition.[2]

100-2—Standard Specification for Marking Shipments by Contractors.[2]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[3]

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Federal specifications, National Military Establishment specifications, and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.—The following publications, of the issue in effect on invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[3]

[2] Applicable only to Army purchases.
[3] Applicable only to Navy purchases.

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(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va.)

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

(Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., may be obtained from the Interstate Commerce Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Material.**—

3.1.1 *Type I (grade A), types II and III.*—Type I (grade A), types II and III magnesium power shall be manufactured from primary magnesium metal containing not less than 99.80 percent metallic magnesium. Mica fillers or other adulterants shall not be used and "bag house dust" shall not be included in the magnesium powder offered for acceptance.

3.1.2 *Type I (grade B).*—Type I (grade B) magnesium powder shall be manufactured from magnesium metal of such purity that the product meets the requirements of this specification. Mica fillers or other adulterants shall not be used and "bag house dust" shall not be included in the magnesium powder offered for acceptance.

3.2 **Form.**—

3.2.1 *Type I.*—Type I magnesium powder shall consist of shavings, turnings, flakes, plates or any combination of these which meet the necessary granulation requirements shown in table II.

3.2.2 *Type II.*—Type II magnesium powder shall consist of oblong, chip-like particles with rounded edges.

3.2.3 *Type III.*—Type III magnesium powder shall consist of granular or spheroidal particles.

3.3 **Chemical composition.** — Magnesium powder shall conform to the requirements shown in table I.

TABLE I.—*Chemical composition*

	Type I		Type II	Type III
	Grade A	Grade B		
	Percent	*Percent*	*Percent*	*Percent*
Material volatile at 105° C. (maximum)	0. 1	0. 1	0. 1	0. 1
Oil and grease (maximum)	. 02	. 02	. 02	. 02
Carbides (maximum)	. 004	. 004	. 004	. 004
Material insoluble in sulfuric acid (maximum)	. 15	------	. 15	. 15
Metallic iron (maximum)	. 05	------	. 05	. 05
Total iron as Fe_2O_3 (maximum)	. 25	------	. 25	. 25
Aluminum as Al (maximum)	------	10	------	------
Free metallic magnesium (minimum)	96	------	96	98
Aluminum plus free metallic magnesium (minimum)	------	96	------	------

3.4 **Granulation.** — Magnesium powder shall be of the fineness shown in table II. All percentages shall be by weight using U. S. Standard sieves conforming to Specification RR-S-366. The powder shall pass through the required sieves readily without balling or the particles clinging together.

3.5 **Apparent density.**—The apparent density of the magnesium powder when determined as specified in 4.4.11 shall be as shown in table III.

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TABLE II.—*Granulation*

U. S. Standard		Granulation 1		Granulation 2		Granulation 3		Granulation 4		Granulation 5		Granulation 6		Granulation 7		Granulation 8		Granulation 9		Granulation 10		Granulation 11		Granulation 12		Granulation 13		Granulation 14		Granulation 15		Granulation 16		Granulation 17	
Sieve No.		Percent		Percent		Percent		Percent		Percent		Percent		Percent		Percent		Percent		Percent		Percent		Percent		Percent		Percent		Percent		Percent	
Through	Retained	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
6																									100								
8																									99								
8	12																							5									
12	20																							50	10								
20									100																								
20	50																							80	50								
25																					100												
30			100		100																												100
30	40	10		7																													
30	50					10																				10							
40											100																						
40	50	45	25	45	25					10																							
50																								5					100				
	50							12																								15	
50	70	50	30	50	30	60	45																			60	45						
50	100								76																								
50	120										80																						
60																	100																
60	80														1																		
70													100																				
70	80	35	15	30	10	30	15					10														30	15						
80		10		15	5																												
	80																				30												
80	100					30	15																			30	15						
80	120												80			15					50	30											
100						15		12							98								100			15					100	15	
	100																											15					
120										10		15					75			100			97										
120	200																				45	25								40			
140																			98														
200																10				90	25							15					
	200																			10				40	15								
325																														50			

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TABLE III.—*Apparent density*

Granulation No.	Density in grams per milliliter	
	Maximum	Minimum
1	0.65	0.55
2	.65	.55
3	.75	.65
4	.625	.45
[1] 5	----------	----------
[1] 6	----------	----------
[1] 7	----------	----------
[1] 8	----------	----------
[1] 9	----------	----------
[1] 10	----------	----------
11	----------	.45
12	----------	.45
13	----------	.70
14	.75	.65
15	----------	.90
16	----------	.90
17	----------	.90

[1] No determination required.

3.6 General.—

3.6.1 *Analysis of metal.*—A chemical analysis of the metal used in the manufacture of magnesium powder shall be furnished with each lot offered for acceptance.

3.6.2. *Description sheets.*—If the material represented in the lot comes from more than one final processing unit, the contractor shall furnish with each lot of material submitted for acceptance a description sheet giving a complete history of its manufacture. If the lot is blended before packing, this operation shall be considered as the final processing unit.

3.7 Workmanship.—

3.7.1 *Process.—*

3.7.1.1 *Types I and II.*—Types I and II magnesium powder shall be manufactured by grinding, stamping, turning, or any combination of these methods which will yield a suitable product.

3.7.1.2 *Type III.*—Type III magnesium powder shall be manufactured by the atomizing process.

3.7.2 Care shall be exercised to avoid inclusion of foreign material during manufacture.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Size of lots.—Unless otherwise specified, the maximum lot size shall be 2,000 pounds.

4.2 Sampling.—A minimum of three containers shall be selected in such a manner as to be representative of the lot. When lots comprise less than three containers, all the containers in the lot shall be selected. If the material represented in the lot comes from more than one final processing unit, the containers selected shall fairly represent the production of each of the final processing units present in the lot. Before sampling, the contents of each selected container shall be thoroughly blended by rolling and turning end for end. By means of a thief, sufficient magnesium powder shall be removed from each selected container to form a primary sample of approximately 1 pound. This sample shall be mixed thoroughly. An 8-ounce portion of the primary sample shall be placed in a tin canister and labeled so that the container from which it was taken can be easily identified. The remaining portions of the primary sample shall be mixed thoroughly in a grounded metal container and quartered to form a composite sample of approximately 1 pound. The composite sample shall be placed in a tin canister which is nearly filled and labeled to show the name of the material, type, grade, and granulation number, manufacturer, plant, contract or order number, lot number, and number of pounds in the lot. Unless otherwise specified, all acceptance tests shall be made on the composite sample representative of the lot (see 6.3). Primary samples shall be held for possible future examination.

4.3 Inspection.—

4.3.1 *Army.*—Inspection shall be made at the place of manufacture in accordance with Specification 50–0–1.

4.3.2 *Navy.*—Inspection shall be at the place of manufacture unless otherwise specified in the contract or order.

4.4 Tests.—Tests shall be conducted in accordance with the following. For Navy purchases, tests will be conducted at a Government laboratory unless otherwise specified.

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4.4.1 *Form.*—Place a small portion of the sample on a glass slide and examine under a 20- to 30-power microscope.

4.4.2 *Material volatile at 105° C.*—Weigh into a tared dish approximately 5.000 gm. of the sample, heat in an oven for 3 hours at 105° C., cool and weigh. Calculate the loss in weight as percent material volatile at 105° C.

4.4.3 *Oil and grease.*—Extract a 50.00-gm. sample with 250 ml. of chemically pure ethyl ether (redistilled over sodium) for 1 hour in a Soxhlet extractor or similar apparatus having an extracting tube of 250-ml. capacity. Evaporate on steam bath to a volume of approximately 50 ml., and then filter the extract, rinsing the flask with three 10-ml. portions of ethyl ether, catching the filtrate and washings in a clean, weighed 150-ml. beaker. Make a blank determination for residue on the ether. Cover the beaker with a ribbed watch glass. Evaporate the ether on a water bath that is constantly at boiling temperature. The total time of heating on the water bath should be approximately 1 hour. Remove from the water bath and dry the outside of the beaker with a clean cloth. Heat in a constant temperature oven at 100° to 105° C. for 30 minutes, cool and weigh. Calculate the gain in weight as percent oil and grease.

Note.—Considerable care shall be exercised through the entire procedure to prevent contamination of the sample or the extract with grease or dust from the hands or other source.

$$\text{Percent oil and grease} = \frac{100\ (A-B)}{W}$$

where

A = gm. of ether extract.
B = gm. of blank.
W = gm. of sample.

4.4.4 *Carbides.*—

4.4.4.1 *Apparatus.*—Use an apparatus similar to that shown on figure 1.

4.4.4.2 *Reagent.*—Dissolve 0.25 gm. of gelatin in hot water. Dilute to 500 ml. and add 500 ml. of 95-percent ethyl alcohol and 1.25 gm. hydroxylamine hydrochloride.

4.4.4.3 *Procedure.*—Add a weighed portion of 50 gm. of magnesium powder to the reaction flask (A). Transfer approximately 50 gm. of cuprous chloride to test tube (B). Pass nitrogen through the apparatus to remove the air and continue the passage of nitrogen throughout the determination. Lower the test tube and add 15 ml. of concentrated NH_4OH and 30 ml. of hydroxylamine hydrochloride reagent. Raise the test tube, insert stopper, add 100 ml. of distilled water from dropping funnel (C) and switch on the heater. Heat the mixture in the reaction flask just to boiling. A red or pink coloration of the cuprous ammonium chloride solution in the test tube indicates that the magnesium sample contains more than 0.004 percent carbides. Should the cuprous ammonium chloride solution become blue, due to the introduction of air, it must be decolorized with successive 5-ml. portions of hydroxylamine hydrochloride reagent until the blue color disappears.

4.4.5 *Material insoluble in sulfuric acid.*—Transfer approximately 5.000 gm. of the sample to a 600-ml. beaker and dissolve in 150 ml. of 1:5 H_2SO_4. Allow to stand until action ceases and all of the magnesium has dissolved. Filter through a tared crucible and wash thoroughly with small portions of hot water, being careful to keep the volume of filtrate and washings at approximately 200 ml. Reserve the filtrate for determination of total iron (see 4.4.7) and dry the residue on the filtering crucible to constant weight at 105° C. Calculate the gain in weight as percent material insoluble in sulfuric acid.

4.4.6 *Metallic iron.*—Using 100 gm. of sample, pass a magnet or an electromagnet through the powder until no more particles are taken up. Brush the particles adhering to the magnet onto a sheet of glazed paper and separate the magnetic from the nonmagnetic by passing the magnet under the paper and drawing the magnetic particles aside. Repeat this procedure to eliminate as much as possible any adhering particles of magnesium. Transfer the magnetic particles onto a balance pan, weigh, and report as metallic iron. In the event that this weight exceeds the requirements for the grade of powder being tested, analyze for iron as follows: Transfer the weighed material to a 250-ml. beaker and add 100 ml. of 5-percent sulfuric acid. Boil until dissolved. Cool. Pass the solution through a Jones reductor, taking the usual precautions to activate the amalgamated zinc. Wash with 50 ml. of 5-percent sulfuric acid and 150 ml. of water.

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Titrate with standard 0.02N potassium permanganate. Calculate the percent metallic iron.

$$\text{Percent metallic iron} = \frac{5.58\,VN}{W}$$

where

V = ml. of $KMnO_4$ solution used.
N = normality of $KMnO_4$ solution.
W = gm. of sample.

4.4.7 *Total iron.*—Transfer the filtrate from the determination of insoluble matter (see 4.4.5) to a 300-ml. Erlenmeyer flask, add a spiral of aluminum wire, cover with a small watch glass and boil for approximately 10 minutes. Cool under a faucet, remove the aluminum spiral and titrate immediately with N/20 $KMnO_4$ solution. Run a blank determination on the aluminum using 200 ml. of 1:1 H_2SO_4. Calculate the percent total iron.

$$\text{Percent total iron (as } Fe_2O_3) = \frac{7.98\,(V-B)N}{W}$$

where

V = ml. of $KMnO_4$ required for sample.
B = ml. of $KMnO_4$ required for blank.
N = normality of $KMnO_4$ solution.
W = gm. of sample.

4.4.8 *Aluminum (type I, grade B).*—Transfer approximately 0.4000 gm. of the sample to a 400-ml. beaker. Cover with a watchglass and add 10 ml. of concentrated H_2SO_4 down the lip of the beaker. Add 50 ml. of distilled water very slowly, taking care to prevent the reaction from proceeding too rapidly. When the reaction has subsided, boil the sample, if necessary, to dissolve the last particles of metal. If there is any residue, add HNO_3 to clear. Filter the solution if necessary, wash and dilute to 250 ml. Add 25 ml. of a saturated solution of NH_4Cl and 5 ml. of methyl red indicator. Heat to boiling and carefully add 1:4 NH_4OH until just neutral. Immediately filter the precipitate on a Whatman No. 40, or equal, filter paper, first placing a little filter paper pulp in the vortex of the paper. Wash once with warm 2-percent NH_4Cl solution. Carefully wash the precipitate back into the original beaker, taking care not to puncture the filter paper. Dissolve the precipitate in the beaker with 10 to 15 ml. of 1:4 HCl, washing down the sides of the beaker with the acid. Heat to boiling and precipitate as before. Filter through the same paper, wash and police the beaker well. Wash several times with warm 2-percent NH_4Cl solution, then wash twice with distilled water, allow the precipitate to drain well, and transfer the precipitate to a tared crucible equipped with cover. Ash the filter paper on a hot plate and ignite in a muffle furnace above 1,000° C. for 1 hour. Cover, cool in a desiccator and weigh as rapidly as possible. Calculate the weight of residue as percent aluminum as follows:

$$\text{Percent aluminum} = \frac{52.91\,A}{W}$$

where

A = gm. of residue.
W = gm. of sample.

4.4.9 *Free metallic magnesium.*—

4.4.9.1 *Apparatus.*—

(*a*) 1 graduated dropping funnel.
(*b*) 1 generating flask.
(*c*) 1 20-inch reflux condenser.
(*d*) 1 glass-stoppered gas washing-bottle containing concentrated H_2SO_4.
(*e*) 1 drying column containing dehydrite or equivalent.
(*f*) 1 combustion furnace having the tube project from the furnace approximately 6 inches on the inlet side and 2½ to 3 inches on the outlet side. The tube shall be filled with copper oxide held in place by a small asbestos plug.
(*g*) 1 absorption tube containing dehydrite or equivalent.
(*h*) 2 guard bottles containing concentrated H_2SO_4.
(*i*) 1 capsule for weighing and introducing the sample. (This may be conveniently made by cutting off a small vial ¾ inch from the bottom.)
(*j*) 1 suction line.

4.4.9.2 *Assembly of apparatus.*—Connect the gas generating flask to the bottom of the dropping funnel using a Y tube having one side open as an air inlet. Connect the lower end of the reflux condenser to the generating flask and the upper end to the gas washing-bottle. Connect the lower end of the drying column to the gas washing-bottle, and the upper end to the

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inlet of the combustion furnace. Connect the lower end of the absorption tube to the outlet of the combustion furnace and the upper end to the two guard bottles connected in series. Connect the outlet of the second guard bottle to the suction line.

4.4.9.3 *Procedure.*—Plug in the furnace, adjust the rheostats to heat the furnace quickly to 600° C. and turn on the water in the reflux condenser. In approximately 1 hour turn on the suction to draw air through the train at the rate of approximately five bubbles per second in the gas washing-bottle. Weigh the absorption tube to obtain the initial weight, reconnect it into the train, draw air through it for 45 minutes, disconnect, cool for 30 minutes, and reweigh. A blank of 0.4 mg. or less is considered satisfactory. Transfer exactly 1.000 gm. of the sample into the capsule and carefully slide it into the generating flask, being careful not to scatter the powder. Reconnect the absorption tube, adjust the rheostats to maintain the furnace at 580° to 620° C. and carefully add approximately 60 ml. of distilled water, enough to cover the lower end of the inlet tube, through the dropping funnel. Place 30 ml. of 30-percent H_2SO_4 in the dropping funnel and begin the decomposition of the sample slowly by adding the H_2SO_4 a drop or two at a time, being careful to note that the bubbles are constantly entering the inlet tube at the rate of approximately five bubbles per second. It is advisable to have on hand a large beaker filled with cold water in which the generating flask may be immersed if the reaction becomes too violent. After 10 to 15 ml. of H_2SO_4 has been added, the reaction is nearly complete and the remainder of the acid may be added rapidly. Gently boil the contents of the flask for 30 minutes, draw air through for an additional 30 minutes, disconnect the absorption tube, cool, and weigh. Disconnect the generating flask and rinse the dropping funnel and the inlet tube well with distilled water to remove acid. Calculate the gain in weight of the absorption tube, due to the water equivalent of the hydrogen generated by the metallic magnesium, as percent free metallic magnesium.

$$\text{Percent free metallic magnesium} = \frac{135\ (A-B)}{W}$$

where

A = gain in weight of absorption tube.
B = gm. of blank.
W = gm. of sample.

4.4.9.4 *Alternate procedure for determination of free metallic magnesium (eudiometer method.)*—When desired, the following eudiometer method may be used: Assemble an apparatus similar to that shown on figure 2. The gas measuring buret shall be graduated in 0.2 ml. The reaction flask shall contain 100 ml. of 3-percent (by volume) of concentrated H_2SO_4 which has been saturated with hydrogen gas. Means shall be provided for dropping the sample into the acid. The weight of sample to be used will depend on the metal content and the temperature and pressure at which hydrogen is measured. At standard conditions of temperature and pressure a 0.47-gm. sample of pure magnesium will yield about 470 ml. of hydrogen. The weight of sample should be roughly computed to give a volume of hydrogen within the capacity of the graduated portion of the gas buret. The system must be in equilibrium with respect to temperature and pressure and checked for any gas leakage before the determination is started. The weighed sample is folded in a piece of paper and placed in position. The temperature is recorded after comparing the reading of the thermometer in the reaction flask with a thermometer located in the close vicinity of the apparatus. The volume is also recorded after the pressure is equalized by means of the leveling bulb. The confining liquid in the pipet is water saturated with hydrogen gas and containing 0.1-percent sulfuric acid by volume and a few drops of methyl orange to facilitate the observation of the solution in the apparatus. Allow the sample to drop into the acid, and shake. After the reaction has ceased, again bring the system to equilibrium by cooling all parts of system to the initial temperature ($\pm 1°$ C.) and again equalizing the pressures. Record the volume of hydrogen gas evolved and calculate to percent free metallic magnesium as follows:

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$$\text{Percent free metallic magnesium} = \frac{R\,\frac{(P-P_w)}{760}\times\frac{273}{(273+T)}\times 0.001086\times 100}{W}$$

$$= \frac{0.03901R(P-P_w)}{W(273+T)}$$

where

B = reading of gas buret in ml.
P = reading of barometer in mm. Hg.
P_w = vapor pressure of water at temperature T.
T = temperature of the system at equilibrium in degrees Centigrade.
W = gm. of sample.

4.4.9.5 *Referee method.*—In case of dispute, the water conversion method shall be used.

4.4.10 *Granulation.*—Place a weighed portion of approximately 50 gm. of the sample on the top sieve of the nest of sieves assembled as specified in table II and provided with a bottom pan. Cover and shake for 30 minutes in a mechanical shaker geared to produce 300±15 gyrations and 150±10 taps of the striker per minute. Weigh the portions retained by each sieve and calculate to a percentage basis as required.

4.4.11 *Apparent density.*—Assemble a Scott volumeter as shown on figure 3 having a No. 8 U. S. Standard sieve in the upper hopper, so that the tared receiving cube, one cubic inch in volume, is directly under the lower funnel and rests on the base of the apparatus. Slowly pour approximately 15 to 20 gm. of the sample into the hopper and brush if necessary through the sieve in order to fill the receiving cube to overflowing. By means of a spatula, carefully strike off the excess magnesium from the cube, taking care not to jar the cube during this operation. Weigh the cube and contents, and from the volume and weight of magnesium calculate its apparent density. Make three tests and calculate the average of the apparent density values as follows:

$$D = \frac{W}{V}$$

where

D = apparent density.
W = gm. of sample.
V = volume of receiving cube in ml.

4.5 **Resubmission.**—If the composite sample representative of the lot fails to pass the inspection tests, the manufacturer shall have the option of having an analysis of each primary sample made at no further expense to the Government. The manufacturer may then remove or replace defective portions of the lot represented by the primary samples which fail to meet the requirements, and resubmit the lot for acceptance, provided that the markings on the containers are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the inspector.

5. PREPARATION FOR DELIVERY

5.1 **Packing.**—Unless otherwise specified in the contract or order, magnesium powder shall be shipped in unlined, full open head steel drums protected against corrosion and conforming to the requirements of Specification 17E of Interstate Commerce Commission Regulations for Transportation of Explosives and other Dangerous Articles, etc. Containers shall contain a maximum of 110 pounds. Each drum shall be provided with a tubular rubber cover gasket, firmly cemented in place. When the jackscrew is tightened, the rim shall be tapped with a mallet to insure complete and proper sealing of the gasket. If a drum is to be reused for the shipment of magnesium powder, new gaskets shall be used each time. In addition, each drum reused shall pass the tests and conditions for new drums.

5.2 **Marking.**—Marking to insure safe handling shall conform to Interstate Commerce Commission Regulations for Transporation of Explosives and Other Dangerous Articles, etc. Unless otherwise specified, each container shall be plainly marked with the following information completed.

Minimum percent free metallic magnesium.
Type of magnesium (flaked, chip, atomized).
Nominal mesh size.

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5.2.1 In addition to any other special marking required by the contract or order, shipments for the Army shall be marked in accordance with Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

6. NOTES

6.1 Intended use.—Magnesium powder covered by this specification is intended for use in the manufacture of tracer, igniter, signal flare and pyrotechnic compositions. Grade B material will ordinarily be manufactured from scrap. Type II material is required for the manufacture of bursting igniters. In general, higher density material will give longer burning times in tracers.

6.2 Ordering data.—Invitations for bids, contracts or other purchasing instruments should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, grade, and granulation required (see 1.2).

6.3 When in the opinion of the inspector, the operation of the final processing units was not uniform during the time the material in the lot was produced, he should require a sieve analysis on each primary sample (see 4.2) or on suitable sublot composite samples.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Other interest:
Army—C.
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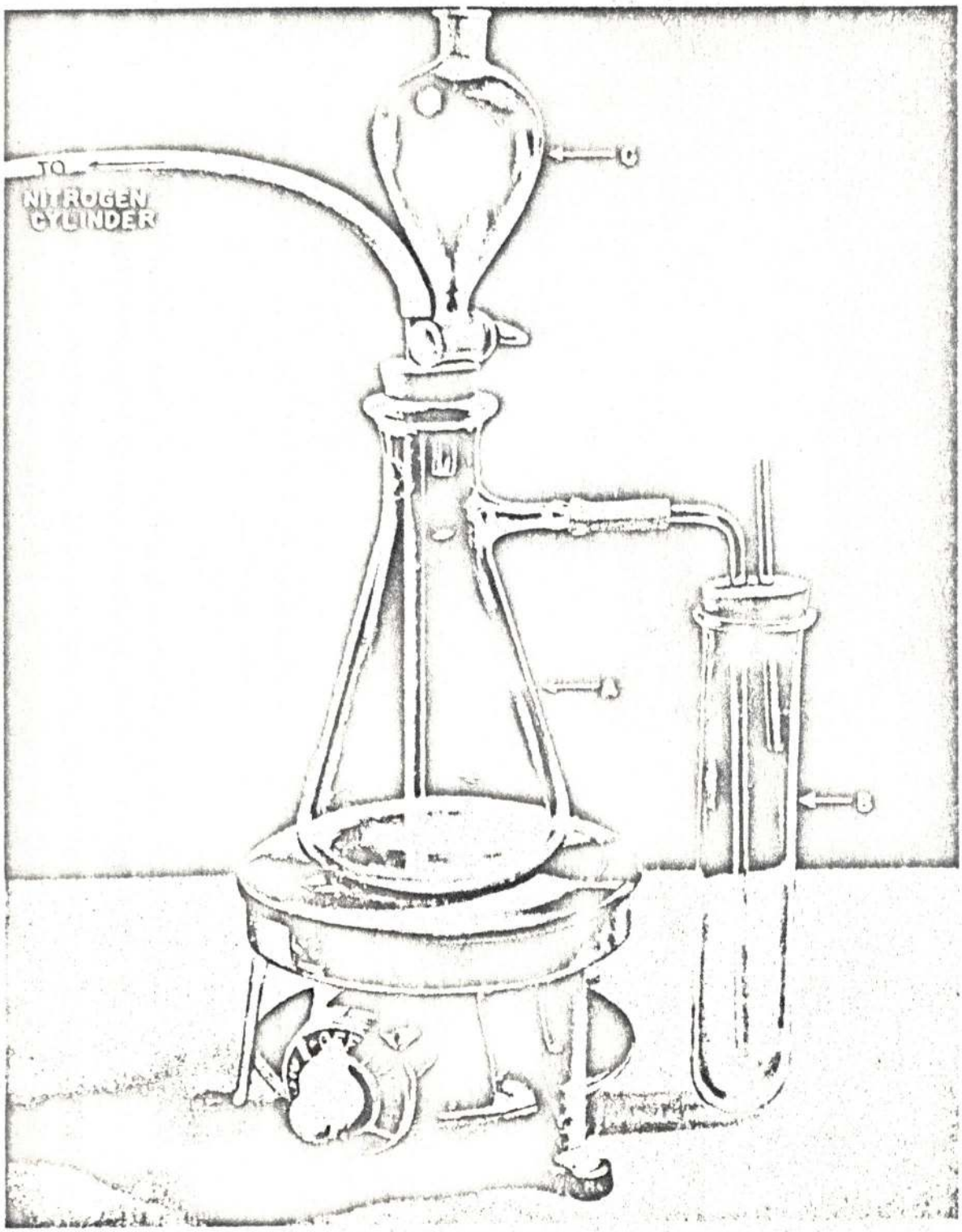


FIGURE 1.—Apparatus for Determination of Carbides in Magnesium.

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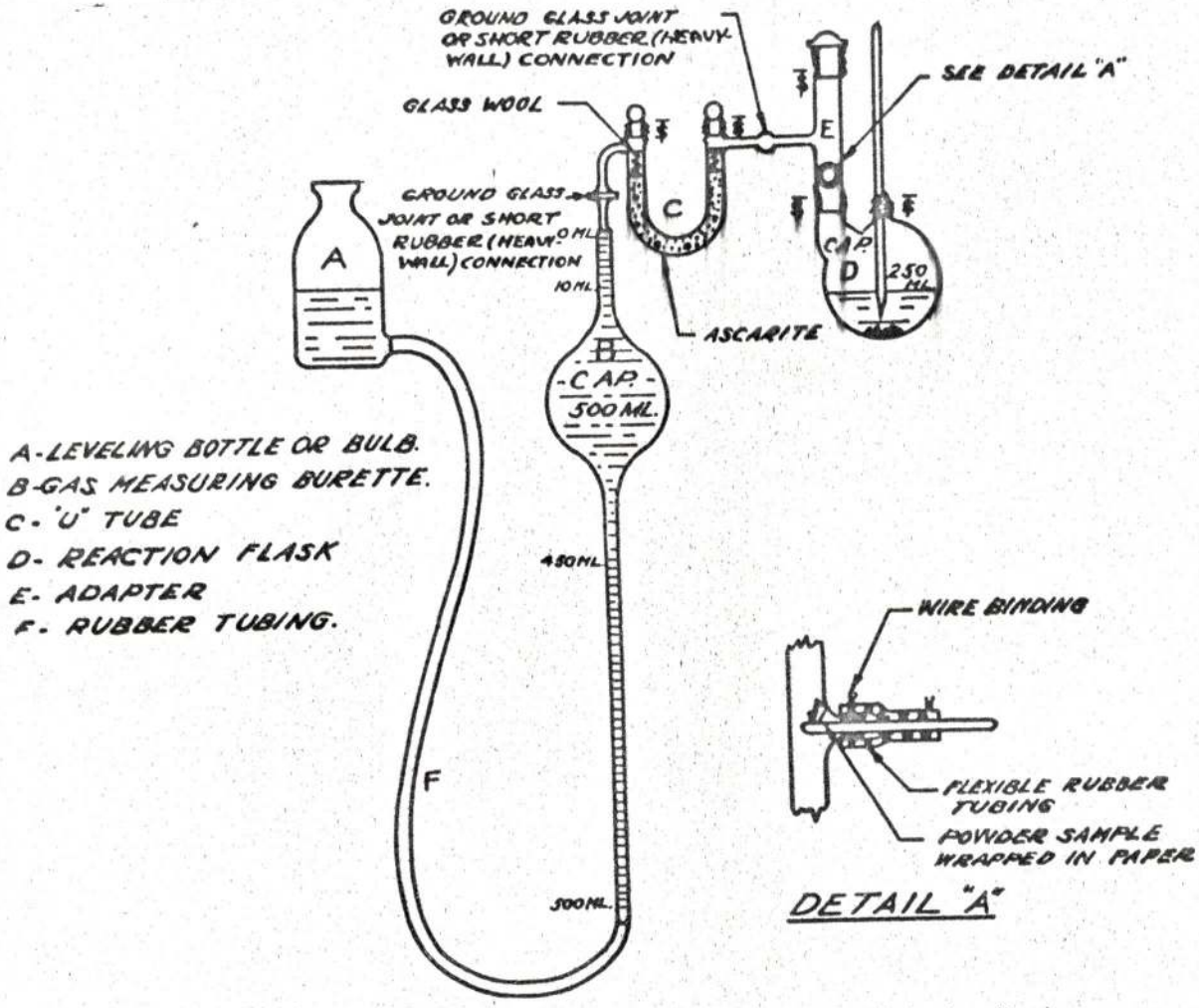


FIGURE 2.—Eudiometer for Determination of Magnesium Metal in Magnesium Powder.

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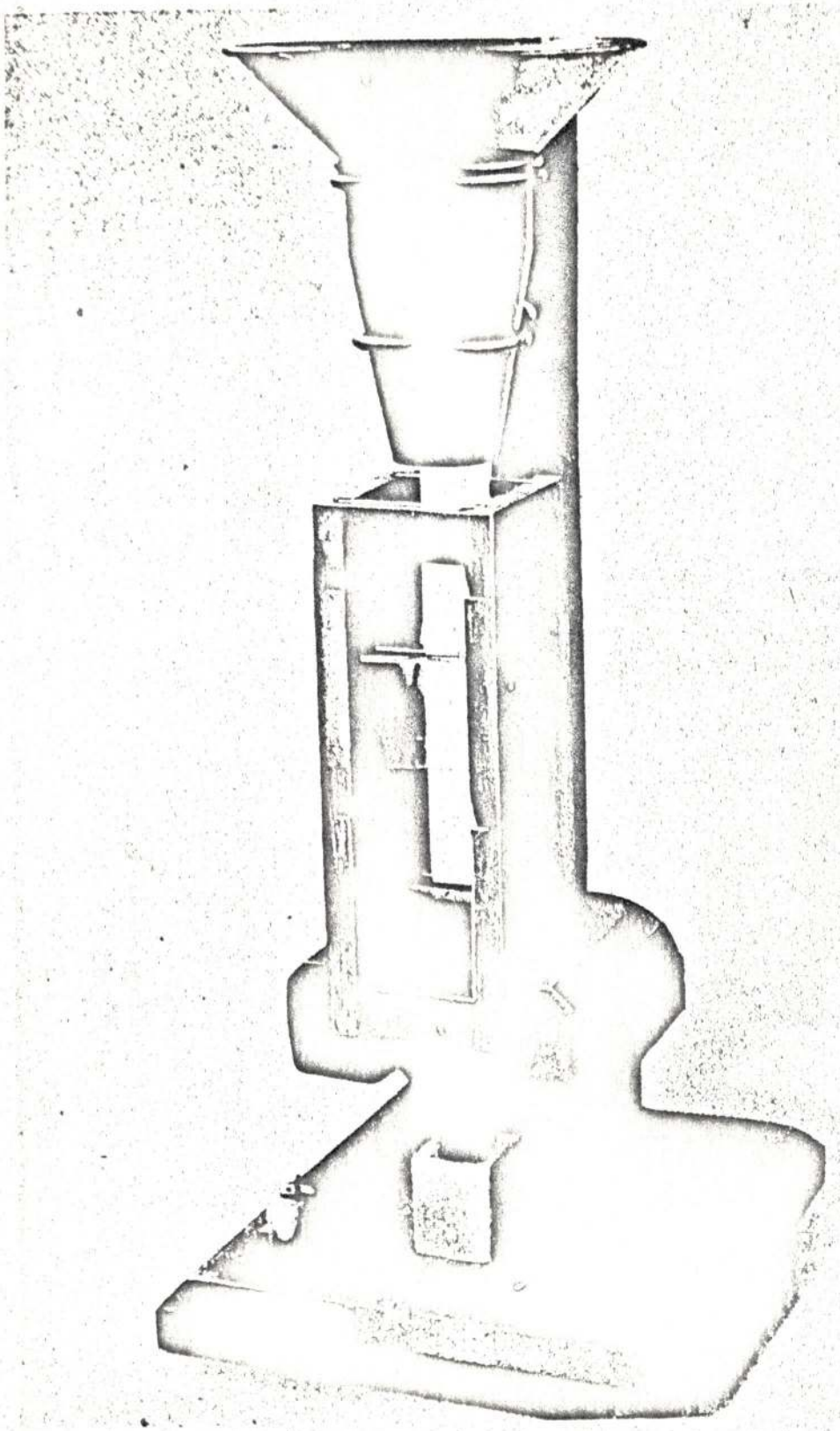


FIGURE 3.—Apparatus for Determination of Apparent Density.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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